

CSC-648-01-SU26

Project: Student Center

Team #4: BigBacks

Version History

Version: Milestone 1 Version 1	Date: 6/14/2026

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Revision History

Date	Member	Task	Details

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Executive Summary

Student Center - with Significant Improvements

Our project idea is a Student Center web application. The goal of this project is to significantly improve upon the student academic planning experience by making common Student Center functions more immediately accessible and easier to understand, while also providing useful features that were otherwise not provided. Instead of redesigning Canvas, which focuses mainly on assignments, grades, and course materials, our project focuses on the administrative and planning aspect of a student's academics. This application will help students view and organize their class schedule, locate class buildings on a campus map, track degree requirements, and receive guidance when planning future courses. This system is especially helpful for new students, transfer students, international students, and students who are close to graduation. By improving the user experience of a student center, our goal is to reduce confusion during registration and help students make better academic decisions.

List of tentative unique features

In this section we will overview why the tentative unique features are unique and why competitors might not be implementing them.

Feature 1: Class Schedule Integration with Campus Map

This feature allows students to view their class schedule, and when prompted, they can be shown the location of the buildings where their classes take place on an integrated campus map. This is especially useful for new students, since they would otherwise need help from other sources or other students to locate these buildings.

Why is it unique? Many student center systems show a student's class schedule, and many university websites provide a campus map, but these tools are usually separated. Our feature combines the class schedule and campus map into one connected experience. Instead of manually searching for each building, students can directly connect their schedule to the physical campus layout.

Competitors in the same domain, such as traditional student portals, usually focus on registration, tuition, grades, and basic schedule information. Campus maps are often provided as separate web pages and are not deeply connected to a student's personal schedule. We believe many existing systems do not implement this because student information systems and campus map systems are often managed separately by different university departments.

Feature 2: Degree Requirement Tree/Roadmap

This feature displays a proper roadmap for students to understand what requirements they will need for graduation. When students select or plan courses, the system will show how those courses relate to their remaining requirements through a more user-friendly display. Requirements can be separated into categories such as major required courses, major elective courses, general education courses, and remaining total credits.

We believe this feature is unique since many students often find class registration already difficult and frustrating because degree requirement and prerequisite information is often spread across degree audit pages, department websites, advising documents, and catalog requirements. Our feature presents this information in a more student-friendly way by introducing a specific format that connects course planning directly into a roadmap for graduation.

Existing competition student center systems may already provide degree audit tools, but they are often difficult to read and not designed around course planning. They may show completed and missing requirements, but they do not always clearly guide students while they are actively selecting future courses. We believe competitors may not implement this in a more user-friendly way because degree requirement rules can be complex, especially with different catalog years, transfer credits, substitutions, and major-specific exceptions.

Feature 3: Advanced Courses and Graduate Planning Suggestions

This feature helps students who are close to completing their undergraduate degree plan ahead. If a student has remaining space in their semester schedule, the system can suggest advanced courses that may support future academic goals, such as graduate school preparation. The system would not guarantee that a course counts toward graduate credit, but it would recommend possible courses for the student to discuss with an academic advisor.

This is unique since most student center systems focus on helping students finish their current degree requirements. Our feature goes one step further by helping students plan ahead about future academic goals while also helping save time. It is designed for students who are close to graduation and want to use their remaining time more strategically.

Competitors may avoid this type of feature because graduate credit policies can be different across universities, departments, and programs. Since approval often depends on university policy and advisor review, many systems may not want to make automatic recommendations. Our approach is different because we frame the feature as planning guidance, not as an official approval system. The system can help students identify possible opportunities while still encouraging them to confirm the details with an advisor.

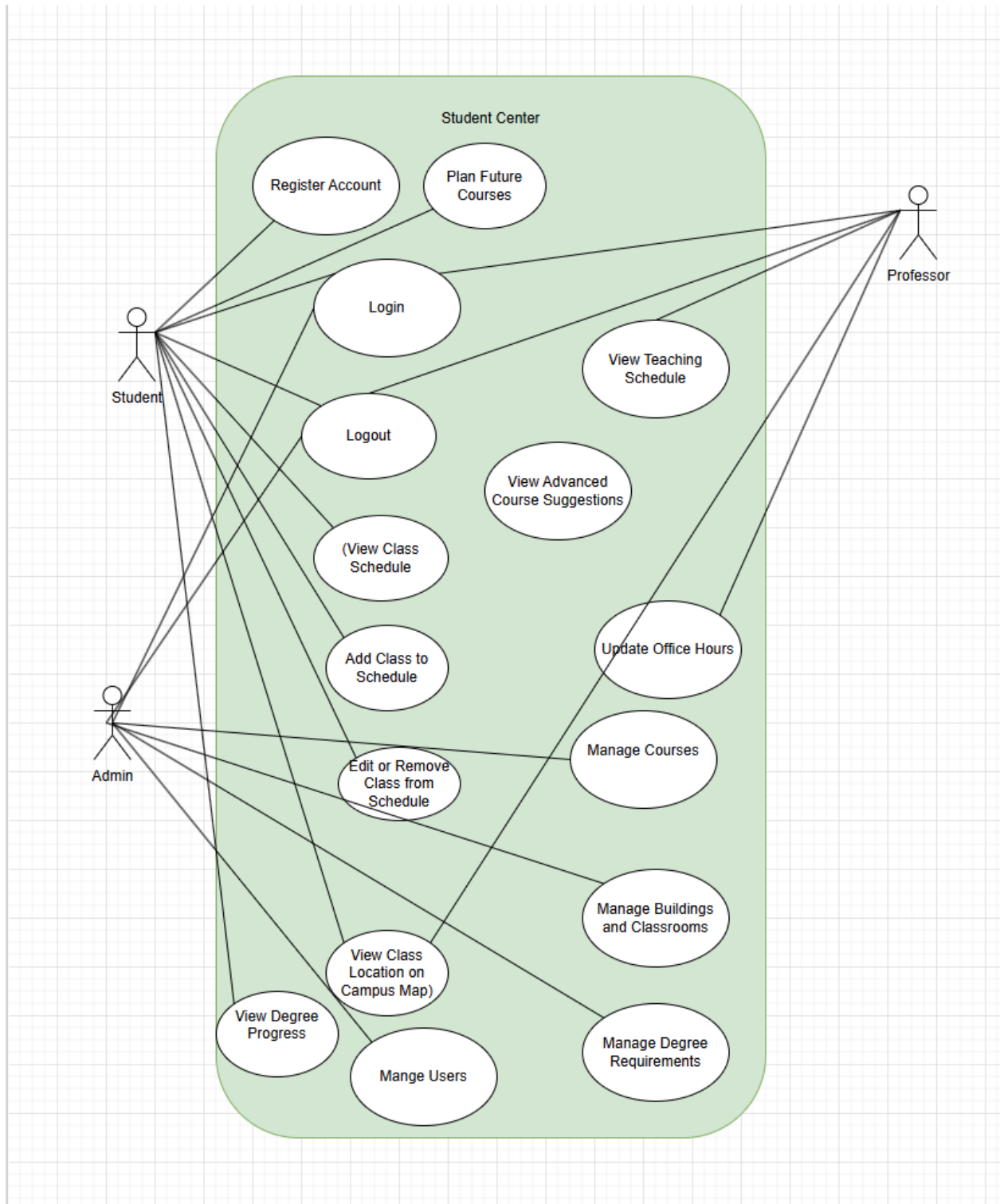
Use Cases

Use Case Name | Actor | Description

1. **Register Account** | **Student** | **A new student creates an account to access the system.**
 - The user does not already have an account.
 - The user has a valid email address.
2. **Login** | **Student / Admin** | **A user logs into the system with their email and password.**
 - The user already has an account.
3. **Logout** | **Student / Admin** | **A user logs out of the system.**
4. **View Class Schedule** | **Student** | **A student views their class schedule, including course name, day, time, building, and room.**
 - The student is logged in.
 - The student has schedule data stored in the system.
5. **Add Class to Schedule** | **Student** | **A student manually adds a course to their schedule on the time the Campus map will highlight the building location for students.**
 - The student is logged in.
 - The student accesses the Academic section.
 - The student manually enrolls in the course.
 - The system adds the class to the student's course schedule and highlights the class building (location) on the Campus map.
6. **Edit or Remove Class from Schedule** | **Student** | **A student updates or removes a class from their schedule.**
 - The student is logged in.
 - The student accesses the Academic section.
 - The student chooses between editing or removing course(s) from their schedule.
7. **View Class Location on Campus Map** | **Student** | **The student clicks a class from their schedule and the system highlights the building location on the campus map.**
 - The student is logged in.
 - The student has at least one class in their schedule.
8. **View Degree Progress** | **Student** | **The student views their completed and remaining degree requirements**
 - The student is logged in.
 - Degree requirement data exists in the system.
 - The student has completed or planned course data.
9. **Plan Future Courses** | **Student** | **A student adds future courses to a planned schedule.**
 - The student is logged in.
 - The student accesses the degree roadmap.
 - The student is able to add future courses for upcoming semesters to their planned schedule.

10. **View Advanced Course Suggestions | Student | The system suggests possible advanced courses for students who are close to graduation and may want to prepare for graduate study.**
 - The student is logged in.
 - The student has degree progress data.
 - Advanced course suggestion data exists in the system.
11. **View Teaching Schedule | Professor | A professor views the courses they are teaching, including time, classroom, and building.**
 - The professor has assigned courses in the system.
12. **View Classroom Location on Map | Professor | A professor views classroom building locations on the campus map.**
 - The professor accesses the course page for their classes.
 - The professor clicks the desired course they wish to see the location of on the Campus map.
13. **Update Office Hours | Professor | A professor updates office hour information.**
 - The professor is logged in.
 - The professor accesses the course page and chooses their desired class.
 - The professor edits office hour information.
14. **Manage Courses | Admin | Admin adds, edits, or deletes course records.**
 - The admin is logged in
 - The admin has permission to manage course data.
15. **Manage Buildings and Classrooms | Admin | Admin manages building names, classroom numbers, and map location data.**
16. **Manage Degree Requirements | Admin | Admin updates major, elective, GE, and credit requirement data.**
17. **Manage Users | Admin | Admin manages student, professor, and admin accounts.**

Use Case Diagram



Main Data Items and Entities

The Better Student Center application uses several main data items and entities to support student planning, enrollment, degree progress, campus navigation, and financial information.

Platform Entities:

Student: Individual with access to the platform, capable of showing all their relevant information and access to tools

Administrator: Individual with access to tools that change the format, layout, and visuals of the website depending on their needs.

Profile: Representation of a student's information in one place.

Notifications: Alerts and notices for a user, which may include course updates or other statistical notices.

Course data: Data that contains the amount of students in the form of a list, which shows student IDs, names, registration date and time and waitlists.

Calendar: Weekly view of a student's schedule for a given semester.

Course registration entities:

Cart: List of a student's currently selected courses prepared for enrollment.

Roadmap: List of a student's graduation roadmap which contains completed courses and potential courses in the future.

Enrollment Display: Custom-built enrollment interface that lets students register and plan for courses quickly.

Course page: Page responsible for showing the information for a given course, complete with instructor and class information.

Finances: Shows the student's financial fees for a given semester, can be coupled with the notifications system.

Functional Requirements

1. User

- a. A User shall be able to log in with their university's Single Sign-On (SSO)
- b. A User shall be able to log out of their account
- c. A User shall be identified as either a student or an admin
- d. A user shall be shown features based on their assigned role.
- e. A User shall be able to add or update contact information.

2. Student

- a. A Student shall have an academic level, such as freshman, sophomore, junior, senior, or graduate.
- b. A Student shall have a student type, such as first-time student, continuing student, transfer student, or international student.
- c. A Student shall be able to search for courses.
- d. A Student shall be able to filter courses.
- e. A Student shall view course details.
- f. A Student shall view enrollment information for the current term.
- g. A Student shall be able to add to or remove courses from a planned schedule.
- h. A Student shall be able to enroll in/add, drop, or withdraw from courses when enrollment is open for their student group.
- i. A Student shall be directed to the appropriate Canvas course page when selecting the Canvas button for an enrolled course.
- j. A Student shall view their planned schedule.
- k. A Student shall view a Degree Progress Tree based on their declared major.
- l. A Student shall be able to update their declared major for major planning purposes.
- m. A Student shall view completed, planned, and remaining degree requirements.
- n. A Student shall view missing prerequisites for planned courses.
- o. A Student shall view class building locations on campus map.
- p. A Student shall view outstanding charges.
- q. A Student shall be redirected to the payment portal to make payments for outstanding charges.
- r. A Student shall be redirected to the financial aid website for application.
- s. A Student shall view their financial aid awards.
- t. A Student shall be able to accept or decline their financial aid awards.
- u. A Student shall view enrollment records from different semesters

3. Admin

- a. An Admin shall be able to add, remove, or update course information.
- b. An Admin shall be able to add, remove, or update class section information.
- c. An Admin shall be able to add, remove, or update building location information.
- d. An Admin shall be able to add, remove, or update degree requirement information.
- e. An Admin shall be able to add, remove, or update student financial information.

4. Enrollment Information

- a. Enrollment Information shall display the date and time when the student's student type is eligible to begin enrollment.
- b. Enrollment Information shall display the enrollment window for the current term.
- c. Enrollment Information shall notify a student when enrollment is available for their student type.
- d. Enrollment Information shall display the add/drop deadline for the current term.
- e. Enrollment Information shall display the withdrawal deadline for the current term.

5. Course

- a. A Course shall have basic course information (code, title, units)
- b. A Course shall be associated with a department
- c. A Course shall have a detailed course description.
- d. A course shall have a course level (lower-division, upper-division, or graduate level)
- e. A course shall have course attributes (General Education areas, repeatable/nonrepeatable, lecture/lab)
- f. A Course shall have prerequisite information.
- g. A Course shall have available class sections for a selected term.
- h. A Course shall be searchable with the class search.
- i. A Course shall be filterable with the class search.

6. Class Section

- a. A Class Section shall have section details (section number, term, instructor name, meeting time, location, enrollment status, and seat availability).
- b. A Class Section shall be addable or removable from a student's planned schedule.
- c. A Class Section shall have a modality (in-person, online, hybrid)
- d. A Class Section shall have a meeting type (synchronous, asynchronous)
- e. A Class Section shall indicate whether it conflicts with class sections already in a Student's planned schedule.

7. Planned Schedule

- a. A Planned Schedule shall display all courses and class sections added by a student.
- b. A Planned Schedule shall display all relevant information for added/enrolled class sections. (meeting days, meeting times, instructor name, building/room, and instruction format for selected class sections.)
- c. A Planned Schedule shall display added/enrolled class sections in a weekly calendar format.
- d. A Planned Schedule shall calculate the total planned units.
- e. A Planned Schedule shall identify time conflicts between added/enrolled class sections.
- f. A Planned Schedule shall calculate estimated walking time between back-to-back in-person class sections.
- g. A Planned Schedule shall warn a student when travel time may not be enough between two in-person classes.

8. Campus Map

- a. A Campus Map shall display campus building locations.
- b. A Campus Map shall display the building location for a selected class section.
- c. A Campus Map shall display classroom location information for a selected class section.
- d. A Campus Map shall display campus resources (library, gym, student center, and health services).
- e. A Campus Map shall display the department building associated with a student's declared major.

9. Degree Progress Tree

- a. A Degree Progress Tree shall display required General Education courses, Major electives, and Major requisites.
- b. A Degree Progress Tree shall display completed requirements, planned requirements, and remaining requirements.
- c. A Degree Progress Tree shall display remaining units needed for graduation.
- d. A Degree Progress Tree shall display prerequisite information for planned courses.
- e. A Degree Progress Tree shall recommend courses that satisfy remaining requirements.
- f. A Degree Progress Tree shall allow a student to view which courses can satisfy remaining requirements (GE, electives).
- g. A Degree Progress Tree shall update when a student updates their planned schedule.
- h. A Degree Progress Tree shall update when a student changes their declared major.
- i. A Degree Progress Tree shall display which requirement category a planned course satisfies.

10. Enrollment Record

- a. An Enrollment Record shall display a student's academic history across semesters.
- b. An Enrollment Record shall display the semester associated with each course.
- c. An Enrollment Record shall display the grade associated with each course.

11. Student Charges

- a. Student Charges shall display charge information (charge type, amount, due date)
- b. Student Charges shall display the total outstanding balance.
- c. Student Charges shall display the estimated amount owed or refund amount after financial aid is applied.
- d. Student Charges shall provide a link to the university payment portal for payment processing.

12. Financial Aid

- a. Financial Aid shall display name, type (loan or award), term, amount, and status (pending/accepted/declined).
- b. Financial Aid shall be applied toward outstanding charges when applicable.
- c. Financial Aid shall provide a link or redirect to the university financial aid application website.

13. Semester

- a. A Semester shall shows a term name, start date, end date, add/drop deadline, withdrawal deadline, and enrollment period.
- b. A Semester shall be able for users to enroll in the class sections.

14. Department

- a. A Department shall have a department name, office location, and contact information.
- b. A Department shall show a list of courses
- c. A Department shall list many majors.

15. Major

- a. A Major shall have a major name, department belong, required units, and catalog year.
- b. A Major shall shows the degree requirements.

16. Degree Requirement

- a. A Degree Requirement shall have a requirement category, required units, and requirement status.
- b. A Degree Requirement shall be satisfied by courses.
- c. A Degree Requirement shall indicate whether the requirement is completed, planned, or remaining.

17. Billing Profile

- a. A Billing Profile shall be provided to students.
- b. A Billing Profile shall store billing contact information and payment-related account information.
- c. A Billing Profile shall display all outstanding charges, payments, refunds, and financial aid credits.
- d. A Billing Profile shall calculate the current outstanding balance.

Nonfunctional Requirements

1. Performance

- 1.1. The system shall display a visual response within three seconds after a user submits a normal request under expected class-project usage.
- 1.2. The login process shall complete within five seconds after valid credentials are submitted.
- 1.3. The class schedule page shall load within three seconds under normal network conditions.
- 1.4. The campus map shall begin displaying within five seconds after a student selects a class location.
- 1.5. Degree progress calculations shall be displayed within five seconds after the student opens the Degree Progress page.
- 1.6. The system shall support at least 25 simultaneous users during prototype testing without becoming unavailable.

2. Reliability

- 2.1. The deployed application shall be available through a public URL during schedule demonstration and grading periods.
- 2.2. The system shall not lose previously saved schedule, degree progress, building, course, or office-hour data after a normal server restart.
- 2.3. The system shall display a clear error message when an operation cannot be completed.
- 2.4. A failed database operation shall not create incomplete or duplicate records.
- 2.5. The system shall confirm successful create, update, and delete operation visually to the user.
- 2.6. The PostgreSQL database shall be backed up at least once per week during active development.

3. Usability

- 3.1. A first-time student user shall be able to locate their schedule, campus map, degree progress, and planning pages within five minutes without formal training.
- 3.2. A student shall be able to open a class location from the schedule in no more than three user actions.
- 3.3. Navigation labels shall use clear terms such as Schedule, Campus Map, Degree Progress, Course Planning, and Profile.
- 3.4. Student, Professor, and Admin dashboard shall display only functions relevant to the user's role.
- 3.5. Required form fields shall be clearly marked, and validation errors shall identify the field that needs correction.
- 3.6. The interface shall remain readable at common laptop resolutions, including 1366 x 768 and 1920 x 1080.
- 3.7. The system shall provide basic responsive behavior for mobile screens with a width of at least 360 pixels.
- 3.8. Degree progress shall be separated visually into major requirements, major electives, general education requirements, and total credits.

4. Security

- 4.1. All public communication between the browser and the deployed applications shall use HTTPS.
 - 4.2. The application shall use TLS 1.3 where supported by the client and server configuration.
 - 4.3. User passwords shall be hashed using bcrypt before being stored in the database.
 - 4.4. Plain-text passwords shall never be stored in the database or application log files.
 - 4.5. The system shall require authentication before allowing access to Student, or Admin dashboards.
 - 4.6. The system shall use role-based authorization to prevent students from accessing Admin functions.
 - 4.7. Authentication tokens shall expire after a defined period of inactivity.
 - 4.8. Database credentials, JWT secrets, and cloud credentials shall be stored in environment variables.
5. **Privacy**
 - 5.1. The system shall collect only the information required for this project, including names, email address, user role, schedule information, completed courses, planned courses.
 - 5.2. Student schedule and academic planning data shall be used only to provide schedule, map, degree progress, and course planning functions.
6. **Compatibility**
 - 6.1. The application shall support the current versions of Google Chrome.
 - 6.2. The frontend shall operate on Windows, macOS, and Linux through a supported web browser.
 - 6.3. The system shall not require users to install browser plug-in.
7. **Hardware and Networking Requirements**
 - 7.1. The application shall be deployed on an AWS Linux cloud instance.
 - 7.2. The cloud instance shall have sufficient resources to run the React frontend, [Node.js/Express](#) backend, Nginx, and the required application services.
 - 7.3. The application shall be accessible through a device with an internet connection and a supported web browser.
 - 7.4. Remote database access shall be limited to authorized user and approved network addresses.
8. **Media Content Requirements**
 - 8.1. Team-member photos on the About page shall use JPEG, PNG.
 - 8.2. Each uploaded team-member photo shall not exceed 2 MB.
 - 8.3. Images shall include meaningful alternative text for accessibility.
9. **Legal, Licensing, and External Requirement**
 - 9.1. The system shall display a disclaimer stating that degree guidance and advanced course suggestions are planning aids and do not replace official academic advising.
 - 9.2. The system shall state that graduate or advanced course suggestions require confirmation under university and department policies.
 - 9.3. Course, degree, and policy information shown in the prototype shall be identified as sample data when it is not obtained from an official source.
10. **Organizational and Development Constraints**

- 10.1.** The frontend shall be developed using JavaScript and React.
- 10.2.** The backend shall be developed using JavaScript, [Node.js](#), and [Express.js](#).
- 10.3.** The system shall use PostgreSQL as its relational database management system.

Competitive Analysis

Competitor Information

Feature/C ompany	SFSU Student Center (Oracle Peoplesoft)	College of San Mateo (Ellucian Banner)	Infinite Campus	Ex Libris campusM	Workday Student
Strengths	It offers a comprehensive set of features, including everything a student might—or might not—need.	Their interface is very clean and does not present users with too much information at first glance.	Has features for students to check grades, assignments, announcements, and graduation reports. Can be used as a standalone classroom-esque application without a third party app like Canvas.	Well optimized mobile interface that has personalizable dashboards, campus map integration/class locations	Students can save schedules and register directly from their academic plans. Detailed class search filters and has schedule-conflict checks.
Weaknesses	The page contains too much information that is sparsely disorganized, and many features that are important to students are hidden or not immediately obvious.	The information provided on their website is too brief, so students may not know where they should go.	Basic implementation of necessary features. Reported to have security concerns	Primarily a mobile app. Web interface not as polished. Features such as registration, class searching, and degree progress depend on other applications.	Unintuitive, uninspiring user interface. Poor implementation of registration system.
Prices	License price: \$245 / FTE Student Software Update License & Support: \$53.90 / FTE Student / year, \$4.49 / FTE Student / month	Not publicly listed; institution-specific contract pricing. Public contract example: Portland Community College approved a 3-year Banner ERP licenses and support contract for \$2,592,481.	Tiered per-student pricing. \$6-20 per student annually. Cloud hosting costs an additional \$1 per student.	Not publicly listed.	Annual subscription price for Workday Student Service: \$166.38 / FTE (before implementation, integrations, consulting, training, or optional products)
Student Enrollment/Class Search	The Enrollment and Class Schedule is not together. Student has to go to other page to Scratch the class they want	On the main page, there is one option called "Register for Classes" and another called "Browse Classes." However, when students open one of these pages, they cannot access	Not available. Infinite Campus is primarily used by K-12 schools where education is continuous for students.	Class search and registration is done through a third party student information system rather than campusM itself.	Comprehensive class search system that directly links with saved schedule or academic plan system.

		the other page directly. As a result, students who want to enroll in classes have to constantly move back and forth between pages in order to browse and register for classes. This makes the process inconvenient and inefficient.			
Weekly Schedule	The Weekly Schedule will provide the information in a weekly calendar, But on the page it won't tell Student where is the Building Location is.	The weekly schedule is located inside the registration information section, which makes it hard for new students to find. Based on interview feedback, some students may not even know this feature exists unless someone tells them.	The weekly schedule is displayed through an extremely simple calendar.	Students can view their schedule by day, week, or year. Class detail is displayed, which can include the instructor, time, and location. Has support for the device's default map application also.	Class schedule is built into the registration and academic planning system. Has block-style course calendar view for students to see where each class occupies in their school day.
Degree Planner	The Degree Planner will provide the information to student, but on the Planner it won't show the Instructor of who is teaching the class for following semester, and if Student delete the course or want to make some change, the Student has to find the Class area by themselves; the System won't automatic provide the information to student	The system does not have a Degree Planner directly integrated into the Student Center. Instead, this function is placed on the school's main website as a separate resource. Because it is not connected to the Student Center, students may need to leave the student portal and search the school website manually to find degree planning information..	Infinite Campus has academic planner and graduation progress features. Students can plan courses across multiple school years and see whether they're meeting graduation requirements. (K-12 graduation requirements only)	Does not feature a built-in degree planner. Degree/schedule planning is done through a third party student information system rather than campusM itself.	Students can view completed, planned, and remaining course requirements and use their academic plan while preparing for registration.

Feature Comparison with Competitors

Feature	Oracle Peoplesoft	Ellucian Banner	Infinite Campus	Ex Libris campusM	Workday Student	Our Student Center
Campus Map Integratio n	-	-	-	++	-	+
Schedule Conflict Detection	-	+	+	-	+	++
Class Search/En rollment	+	+	-	-	++	++
Class-Wal king Time Calculatio n	-	-	-	-	-	+
Degree Progress Report	+	-	+	-	++	++
Weekly Schedule View	+	+	+	+	++	++

Project Readiness Checklist

The team (Big Backs) has established regular communication channels and a weekly meeting schedule. We meet in-person on Tuesdays and Thursday before/during class and the team lead hosts an asynchronous Discord meeting on Friday afternoon. An overview of the team meeting is posted after it concludes for team members who were not able to attend. Discord serves as the main platform for live discussion as well as ongoing communication.

Project responsibilities have been divided among team members, and the assigned roles are listed on the title page. Assigned roles may be changed during the course of the semester.

The team has finalized our technology stack and has started reviewing the tools needed for our application development. In addition, the repository has been organized to meet Milestone 1 requirements. Our technology stack is listed in the next section.

High Level Architecture and Technologies

Components in **bold text** are part of the instructor-approved tech stack.

Component	Configuration
Cloud provider	Amazon Web Services
SSL Certificate	AWS Certificate Manager
Application Server	AWS EC2
Operating System	Ubuntu Server
SSH Username	ubuntu
SSH Port	22
Database Service	Amazon RDS for PostgreSQL
Database Port	5432
Database Name	better_student_center
RDS endpoint	
Frontend	React with Vite
Backend	Node.js with Express.js
Web Server	Nginx
Repository Location on EC2	/home/ubuntu/csc648-848-su26-project-Big Backs-04
Public Frontend Directory	/var/www/html
SSL / HTTPS	HTTPS using TLS 1.3

Team Familiarity with Tech Stack

Team Familiarity Table

Team Member	Javascript	React	Node.js	PostgreSQL	AWS	TLS 1.3	average
Wen Chien Yen	3	3	3	2	3	0	2.33
Banghao Yuan	3	0	3	0	0	0	1.00
Sean Wang	3	1	3	2	0	0	1.50
Oscar Wang	4	0	3	1	0	0	1.33
Enoch Lin	3	1	3	2	0	0	1.50
Kashif Zada	2	1	3	1	0	0	1.17

We scored the major technologies that are central to our project stack. Smaller supporting libraries such as React Router, Axios, bcrypt, JWT, dotenv, Leaflet.js, and Chart.js are listed in the frameworks/libraries section, but they are not scored individually because they are supporting tools rather than the main technology stack. Our team will learn and apply these libraries as needed during implementation.

Overall Team Familiarity Summary

The team has the strongest familiarity with JavaScript and Node.js, which supports our decision to use JavaScript for both frontend and backend development. React familiarity is lower for some members, so frontend tasks may require additional learning and support. PostgreSQL familiarity is moderate to low, but it is still manageable for the project because the database design will focus on relational data such as users, schedules, courses, buildings, and degree requirements. AWS and TLS 1.3 familiarity are currently low, so deployment and security configuration may require extra preparation and documentation.

Team Contributions

Team lead assigned contribution scores. All team members have submitted confidential feedback via the instructor provided Google form.

Team Member	Wen Chien Yen	Oscar Wang	Banghao Yuan	Sean Wang	Enoch Lin	Kashif Zada
Contribution Score	7	6	6	5	5	4